Exercise 3

Problem 1 (Multiple-Choice Exam)**.** The final exam of the Lecture is a multiple-choice exam with 20 questions. Each question has 4 possible answers of which exactly one is correct. Assuming you only know 8 answers out of the 20 questions, and therefore you choose the remaining answers at random. Let X be your final score to the exam, counting the nr. of correct answers.

1. Compute the probability of passing the test, that is the event of $X \geq 10$.
2. Compute the probability of getting the grade 2 on the exam, that is $15 \leq X < 18$.

Use the probability mass function

$$P(y) = \binom{n}{y} p^y (1-p)^{n-y}$$

of the *Binomial distribution*, which describes the probability of successes $y \in \{0, 1, 2, \ldots, n\}$ out of n tries with success probability $p \in (0, 1)$.

Problem 2 (Airplanes)**.** Suppose that an airplane engine will fail, when in flight, with probability $1 - p$ independently from engine to engine; suppose that the airplane will make a successful flight if at least 50% of its engines remain operative. For what values of p is a four-engine plane preferable to a two-engine plane?

Problem 3 (Shoplifting)**.** A security guard in a shop is responsible for preventing shoplifting. If a customer attempts to steal something, the guard has an 80% chance of noticing the attempt and stops the perpetrator. However, there is also a 1% chance that the guard mistakenly stops an honest customer for inspection.

If you are in the shop and observe another customer being stopped for inspection, what is the probability that this customer was stopped mistakenly, assuming that only one in one thousand customers are thieves?

Problem 4 (Triangular Inequality)**.** Let A, B and C be arbitrary events. Prove the following inequality;

$$P(A \,\Delta\, C) \leq P(A \,\Delta\, B) + P(B \,\Delta\, C).$$

Problem 5 (Switch?)**.** A researcher is testing the reasoning capabilities of a new robot. The robot has a very low charged battery and must choose from five available batteries. It is told that only one of these batteries is charged, allowing it to avoid running out of power.

Initially, the robot has no information and selects one of the five batteries with equal probability. After this selection, the researcher removes three of the four unselected batteries, informing the robot that only empty batteries are being removed.

Now, the robot faces a final choice: it can either stick with its original selection or switch to the other remaining battery.

What should the robot do, and why?

Problem 6 (Crunching Numbers – *Bonus Problem*)**.** Perform a computer simulation using `R` for Problem 5. Implement the random experiment in `R` to execute both possible strategies the robot could take and determine the better option.

Repeat the random experiment at least 1 000 times to draw your conclusions.